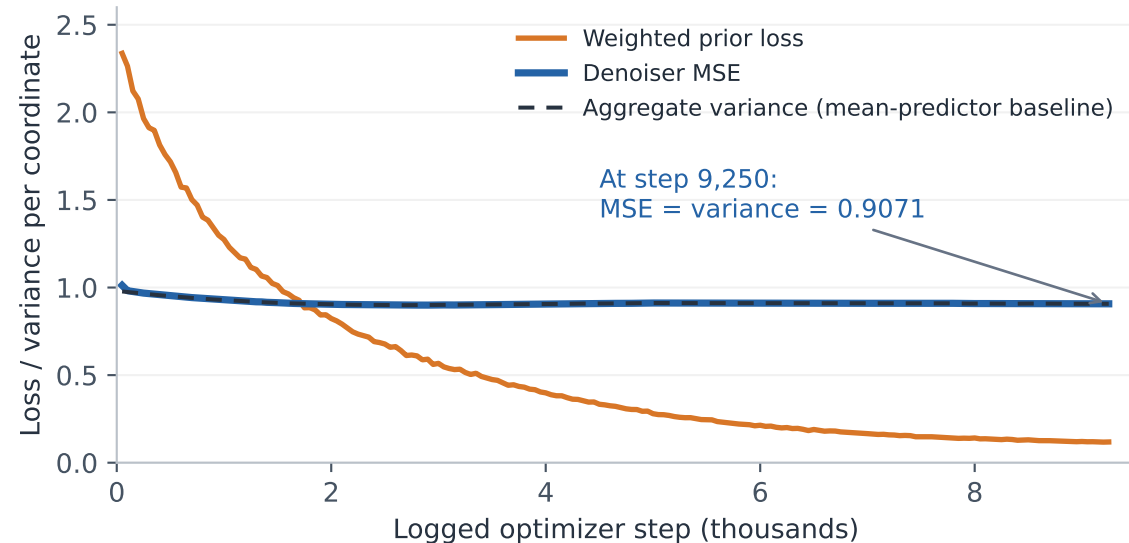


Denoising diagnostics: objective progress is not language progress

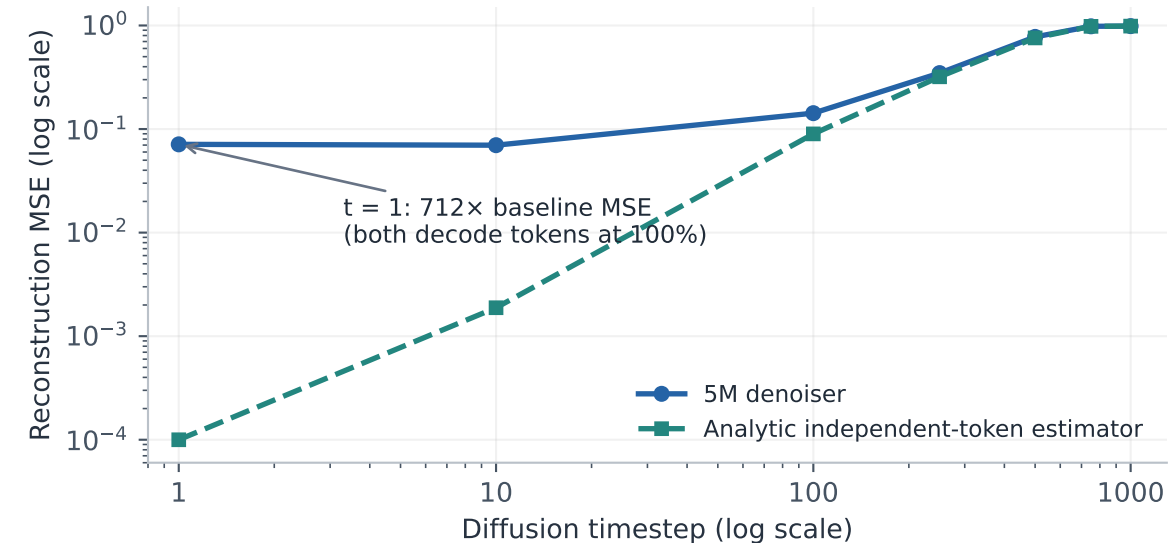
5M checkpoint: 10,000 optimizer steps | CPU reconstruction and paired context tests

Training-distribution diagnostics, not held-out evaluation. 10M panel uses a legacy log with source mismatch.

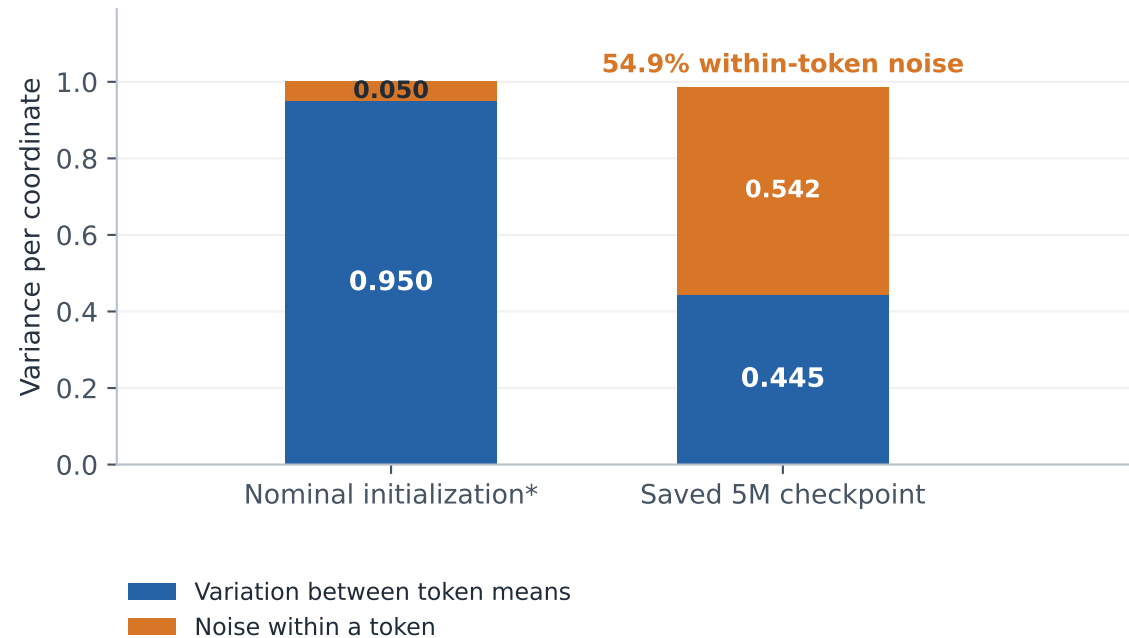
A 10M legacy run: the prior improves; denoising stalls



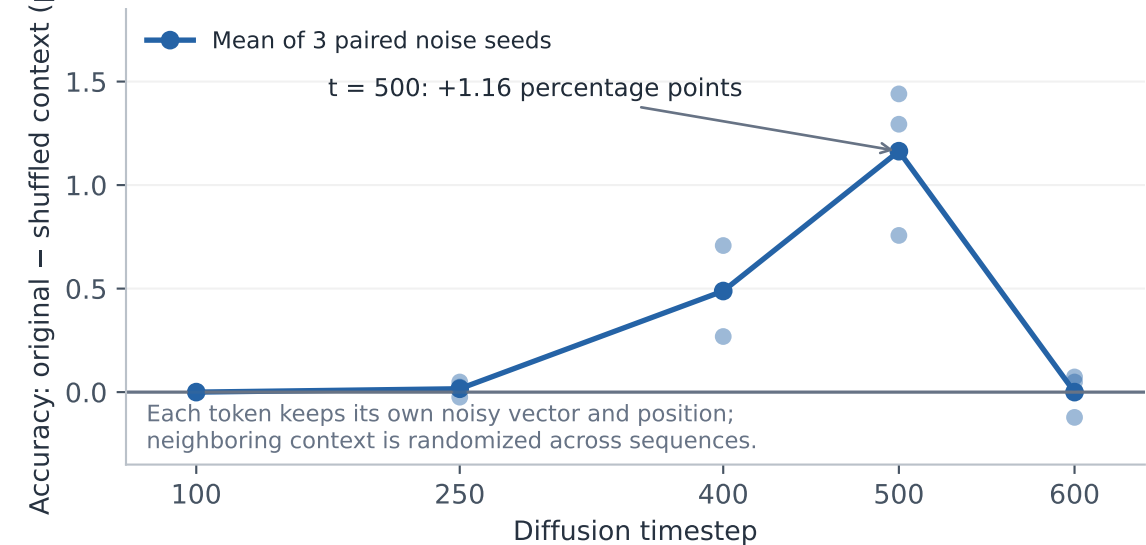
B 5M model trails a baseline that uses no context



C 5M embeddings allocate over half their variance to noise



D 5M context helps only modestly in this diagnostic



* Nominal initialization uses mean-coordinate variance 0.95 and within-token variance 0.05; the initial corpus-weighted between-token variance was not measured. Checkpoint variance is weighted by corpus token frequencies.

B: one 2,048-token diagnostic batch. D: same 4,096 tokens across 3 noise seeds; dots are seed results, not confidence intervals.